# Bihar Engineering University, Patna

## B.Tech 4th Semester Examination, 2024

*Course : B.Tech* | *Time: 03 Hours*

*Code : 105402* | *Subject: Design and Analysis of Algorithms* | *Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

***Q.1 Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following is TRUE?
I. Quicksort runs in Θ(n2) time
II. Bubblesort runs in Θ(n2) time
III. Mergesort runs in Θ(n) time
IV. Insertion sort runs in Θ(n) time
(i) I and II only (ii) I and III only
(iii)II and IV only (iv) I and IV only

(b) Worst-case complicity of heap sort is –
(i) O(log n) (ii) O(n2)
(iii) O(n log n) (iv) O(n)

(c) The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are:
(i) Θ(n log n), Θ(n log n) and Θ($n^2$) (ii) Θ($n^2$), Θ($n^2$) and Θ(n Log n)
(iii) Θ($n^2$), Θ(n log n) and Θ(n log n) (iv) Θ($n^2$), Θ(n log n) and Θ($n^2$)

(d) The minimum number of comparisons required to determine if an integer appears more than n/2 times in a sorted array of n integers is
(i) O(n) (ii) O(log n)
(iii) O(n log n) (iv) O(1)

(e) What is the minimum number of stacks of size n required to implement a queue of size n
(i)one (ii) two
(iii)three (iv) four

(f) Dijkstra's algorithm is used to solve _____ problems?
(i) Single source shortest path (ii) All pair shortest path
(iii) Sorting (iv) Searching

(g) The approach followed in Floyd Warshall's algorithm is
(i) Greedy approach (ii) Dynamic approach
(iii) Backtracking (iv) Divide and conquer

(h) Hamiltonian path problem is _____?
(i) NP Problem (ii) NP Complete
(iii) P Class Problem (iv) N Class Problem

(i) Which of the following algorithms are used to find the shortest path from a source node to all other nodes in a weighted graph?
(i) BFS (ii) Prim's Algorithm
(iii) Djikstra's Algorithm (iv) Kruskal's Algorithm

(j) Kruskal's Algorithm for finding the Minimum Spanning Tree of a graph is a kind of a?
(i)DP Problem (ii) Greedy Algorithm
(iii) Adhoc Problem (iv) None of the above

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**Q.2** (a) Define Asymptotic Notation. Show that $n^2 + 3\log n = O(n^3)$ [7]

(b) Solve the recurrence relation $T(n) = 2T(n/2) + O(n)$ [7]

**Q.3** (a) Write Divide – And – Conquer Quick sort algorithm and analyze the algorithm for average time complexity. [7]

(b) Simulate Quick sort algorithm for the following example 25,36,12,4,5,16,58,54,24,16,9,65,78 [7]

**Q.4** (a) Construct the max heap for the following: 35 33 42 10 14 19 27 44 26 31 [7]

(b) Explain Strassen's matrix multiplication and its time complexity. [7]

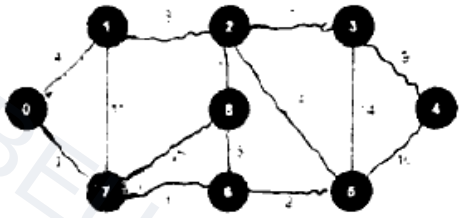
**Q.5** (a) State Huffman Tree. Explain with examples. [7]

(b) What are NP- hard and NP-complete problems? [7]

**Q.6** (a) How 8-Queen's problem can be solved using back tracking and explain with an example. [7]

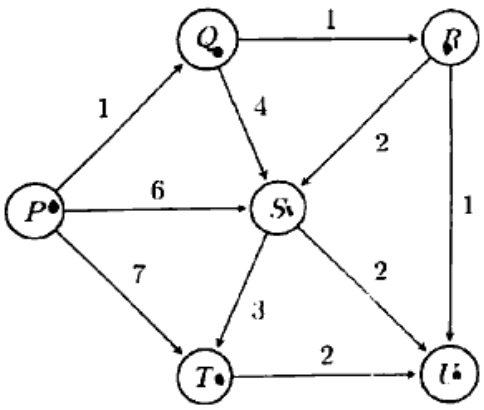
(b) Distinguish greedy method and dynamic programming [7]

**Q.7** (a) Construct minimum cost spanning tree using Kruskal's algorithm [7]



(b) Define spanning tree and minimum spanning tree [7]

**Q.8** (a) Apply single source shortest path algorithm for the following graph [7]




(b) Explain Knapsack Problem. [7]

**Q.9** Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Cook's Theorem

(b) Topological sorting

(c) Travelling Salesman Problem

(d) Masters' theorem



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